

Foundation Mathematics 1017SCG
Week 7 Workshop Solutions

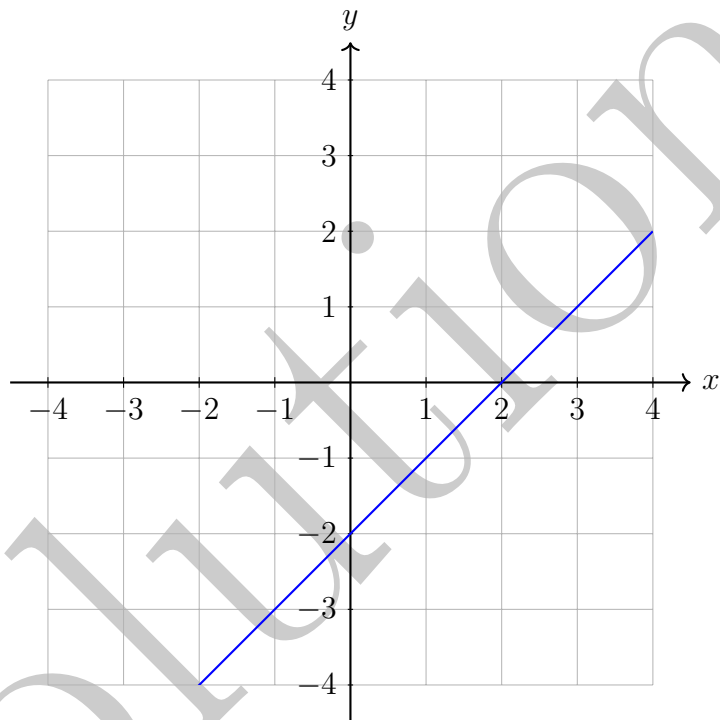
Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a) We have been asked to graph $y = x - 2$.

To find the x -intercept, let $y = 0$.

$$\begin{aligned}x - 2 &= 0 \\x - 2 + 2 &= 0 + 2 \\x &= 2\end{aligned}$$

Therefore the gradient is 1, the x -intercept is $x = 2$ and the y -intercept is $y = -2$.

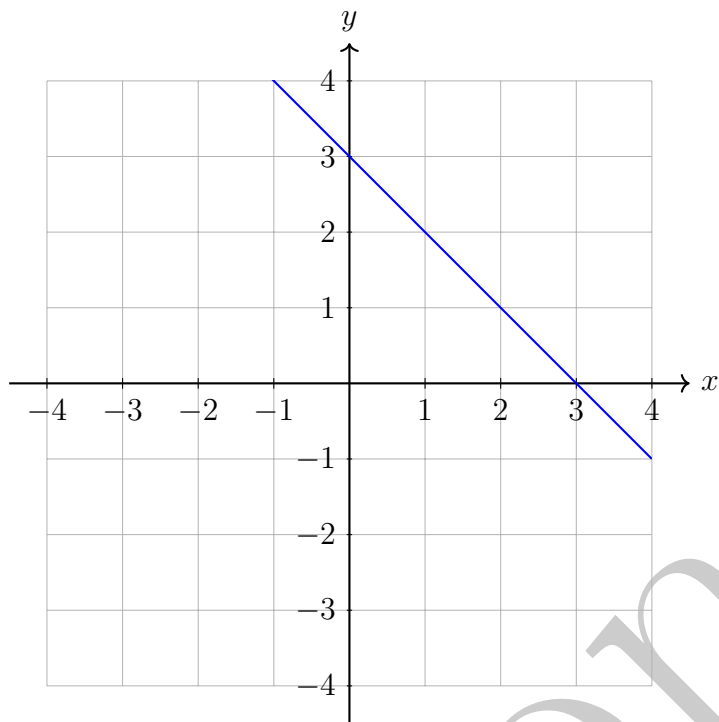


- (b) We have been asked to graph $y = -x + 3$.

To find the x -intercept, let $y = 0$.

$$\begin{aligned}-x + 3 &= 0 \\-x + 3 - 3 &= 0 - 3 \\-1x &= -3 \\\frac{-1x}{-1} &= \frac{-3}{-1} \\x &= 3\end{aligned}$$

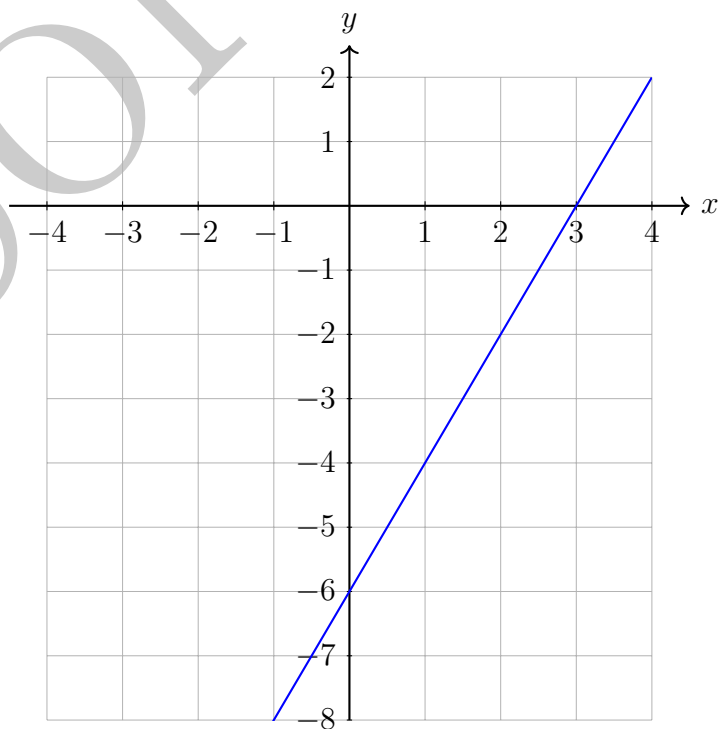
Therefore the gradient is -1, the x -intercept is $x = 3$ and the y -intercept is $y = 3$.



- (c) We have been asked to graph $y = 2x - 6$.
To find x -intercept, let $y = 0$.

$$\begin{aligned}
 2x - 6 &= 0 \\
 2x - 6 + 6 &= 0 + 6 \\
 2x &= 6 \\
 \frac{2x}{2} &= \frac{6}{2} \\
 x &= 3
 \end{aligned}$$

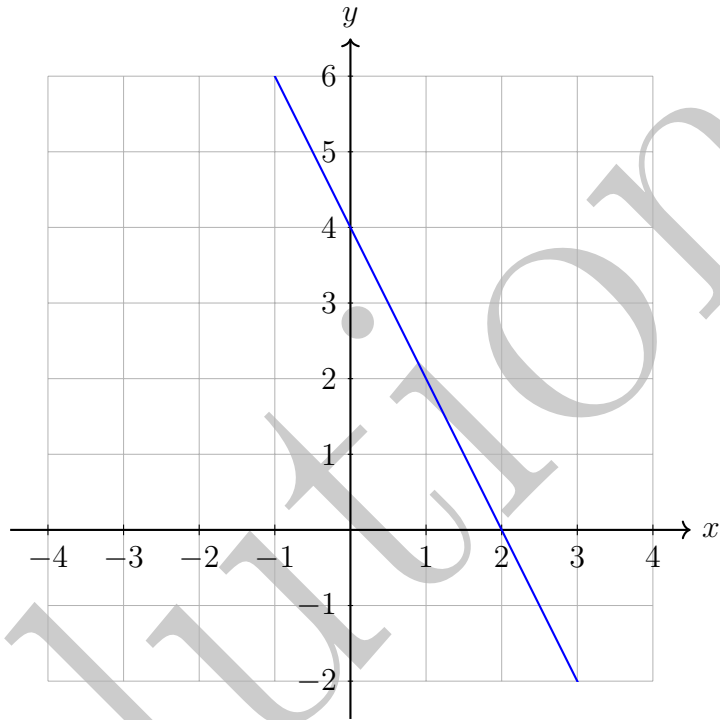
Therefore the gradient is 2, the x -intercept is $x = 3$ and the y -intercept is $y = -6$.



- (d) We have been asked to graph $y = -2x + 4$.
To find x -intercept, let $y = 0$.

$$\begin{aligned}-2x + 4 &= 0 \\ -2x + 4 - 4 &= 0 - 4 \\ -2x &= -4 \\ \frac{-2x}{-2} &= \frac{-4}{-2} \\ x &= 2\end{aligned}$$

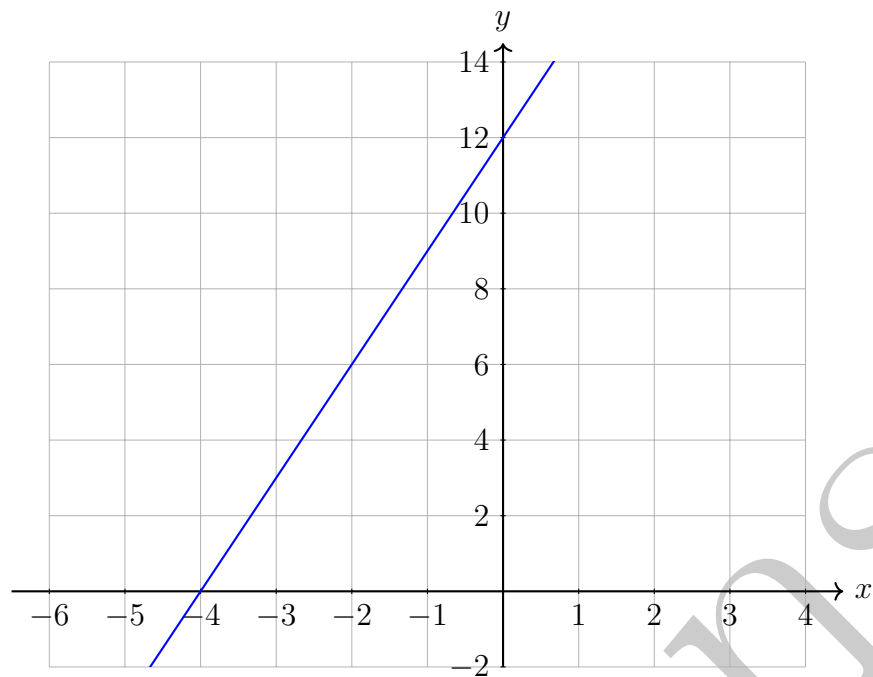
Therefore the gradient is -2, the x -intercept is $x = 2$ and the y -intercept is $y = 4$.



- (e) We have been asked to graph $y = 3x + 12$.
To find x -intercept, let $y = 0$.

$$\begin{aligned}3x + 12 &= 0 \\ 3x + 12 - 12 &= 0 - 12 \\ 3x &= -12 \\ \frac{3x}{3} &= \frac{-12}{3} \\ x &= -4\end{aligned}$$

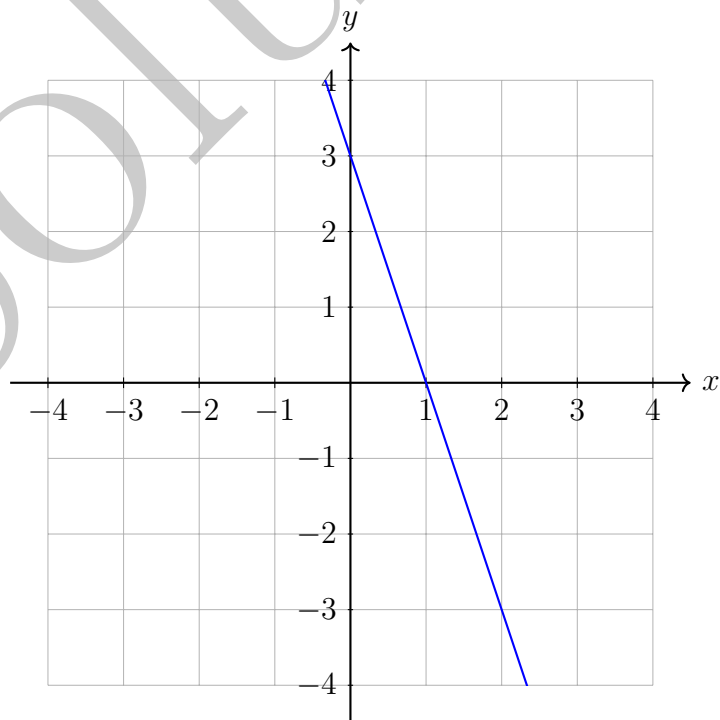
Therefore the gradient is 3, the x -intercept is $x = -4$ and the y -intercept is $y = 12$.



(f) We have been asked to graph $y = -3x + 3$. To find x -intercept, let $y = 0$.

$$\begin{aligned}
 -3x + 3 &= 0 \\
 -3x + 3 - 3 &= 0 - 3 \\
 -3x &= -3 \\
 \frac{-3x}{-3} &= \frac{-3}{-3} \\
 x &= 1
 \end{aligned}$$

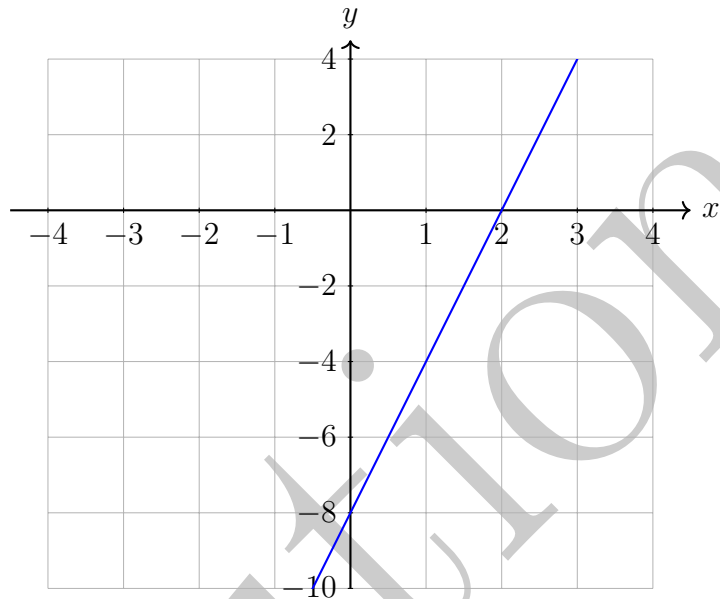
Therefore the gradient is -3, the x -intercept is $x = 1$ and the y -intercept is $y = 3$.



(g) We have been asked to graph $y = 4x - 8$. To find x -intercept, let $y = 0$.

$$\begin{aligned}4x - 8 &= 0 \\4x - 8 + 8 &= 0 + 8 \\4x &= 8 \\\frac{4x}{4} &= \frac{8}{4} \\x &= 2\end{aligned}$$

Therefore the gradient is 4, the x -intercept is $x = 2$ and the y -intercept is $y = -8$.

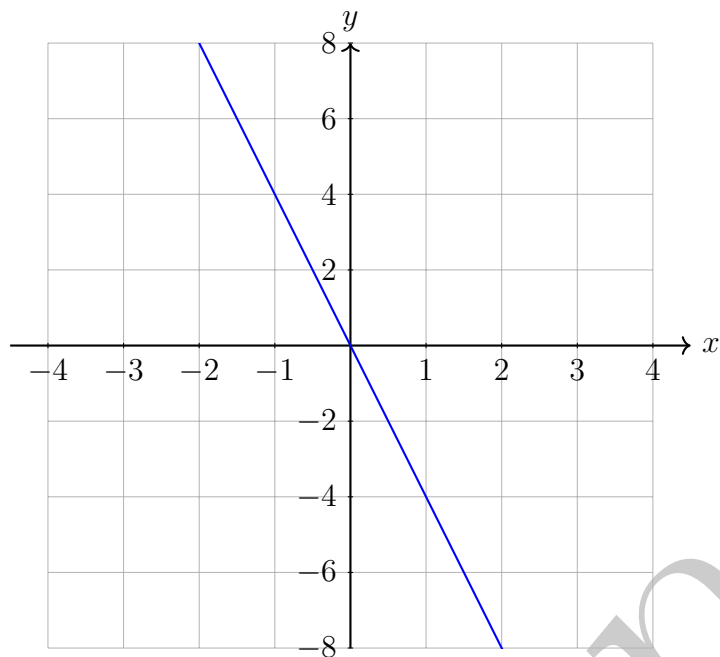


(h) We have been asked to graph $y = -4x$. You may find it helpful to rewrite this equation as $y = -4x + 0$. To find x -intercept, let $y = 0$.

$$\begin{aligned}-4x &= 0 \\\frac{-4x}{-4} &= \frac{0}{-4} \\x &= 0\end{aligned}$$

Therefore the gradient is -4, the x -intercept is $x = 0$ and the y -intercept is $y = 0$.

You may find it helpful to find another point the graph passes through to assist with drawing the graph. For example, substituting $x = 1$ into the equation gives $y = -4$. Therefore, the graph passes through the point $(1, -4)$.



2. (a) We have been asked to find the equation of the line that passes through $(-1, 4)$ and has a gradient of 3.

$$\begin{aligned}
 y - y_1 &= m(x - x_1) \\
 y - 4 &= 3(x - -1) \\
 y - 4 &= 3(x + 1) \\
 y - 4 &= 3x + 3 \\
 y - 4 + 4 &= 3x + 3 + 4 \\
 y &= 3x + 7
 \end{aligned}$$

- (b) We have been asked to find the equation of the line that passes through $(2, -3)$ and has a gradient of -1.

$$\begin{aligned}
 y - y_1 &= m(x - x_1) \\
 y - -3 &= -1(x - 2) \\
 y + 3 &= -1(x - 2) \\
 y + 3 &= -x + 2 \\
 y + 3 - 3 &= -x + 2 - 3 \\
 y &= -x - 1
 \end{aligned}$$

- (c) We have been asked to find the equation of the line that passes through $(-5, 2)$ and has a gradient of 4.

$$\begin{aligned}
 y - y_1 &= m(x - x_1) \\
 y - 2 &= 4(x - -5) \\
 y - 2 &= 4(x + 5) \\
 y - 2 &= 4x + 20 \\
 y - 2 + 2 &= 4x + 20 + 2 \\
 y &= 4x + 22
 \end{aligned}$$

- (d) We have been asked to find the equation of the line that passes through the points $(-3, 1)$ and $(4, 15)$. We will start by finding the gradient, m , of the line.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{15 - 1}{4 - -3} \\ &= \frac{14}{7} \\ &= 2 \end{aligned}$$

Now we can find the equation of the line that passes through the two points.

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 1 &= 2(x - -3) \\ y - 1 &= 2(x + 3) \\ y - 1 &= 2x + 6 \\ y - 1 + 1 &= 2x + 6 + 1 \\ y &= 2x + 7 \end{aligned}$$

- (e) We have been asked to find the equation of the line that passes through the points $(-4, 0)$ and $(0, 3)$. We will start by finding the gradient, m , of the line.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{3 - 0}{0 - -4} \\ &= \frac{3}{4} \end{aligned}$$

Now we can find the equation of the line that passes through the two points.

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 0 &= \frac{3}{4}(x - -4) \\ y &= \frac{3}{4}(x + 4) \\ y &= \frac{3}{4}x + 3 \end{aligned}$$

- (f) We have been asked to find the equation of the line that passes through the points $(1, 1)$ and $(5, 17)$. We will start by finding the gradient, m , of the line.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{17 - 1}{5 - 1} \\ &= \frac{16}{4} \\ &= 4 \end{aligned}$$

Now we can find the equation of the line that passes through the two points.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 1 &= 4(x - 1) \\y - 1 &= 4x - 4 \\y - 1 + 1 &= 4x - 4 + 1 \\y &= 4x - 3\end{aligned}$$

- (g) We have been asked to find the equation of the line that has an x -intercept at $x = 2$ and a y -intercept at $y = 6$. Therefore, the line passes through the points $(2, 0)$ and $(0, 6)$. We will start by finding the gradient, m , of the line.

$$\begin{aligned}m &= \frac{y_2 - y_1}{x_2 - x_1} \\&= \frac{6 - 0}{0 - 2} \\&= \frac{6}{-2} \\&= -3\end{aligned}$$

Now we can find the equation of the line that passes through the two points.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 0 &= -3(x - 2) \\y &= -3x + 6\end{aligned}$$

- (h) We have been asked to find the equation of the line that has an x -intercept at $x = -5$ and a y -intercept at $y = -5$. Therefore, the line passes through the points $(-5, 0)$ and $(0, -5)$. We will start by finding the gradient, m , of the line.

$$\begin{aligned}m &= \frac{y_2 - y_1}{x_2 - x_1} \\&= \frac{-5 - 0}{0 - -5} \\&= \frac{-5}{5} \\&= -1\end{aligned}$$

Now we can find the equation of the line that passes through the two points.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 0 &= -1(x - -5) \\y &= -1(x + 5) \\y &= -x - 5\end{aligned}$$

3. There are several ways that the equation of the line could be determined from the graph.
- Determine the gradient and y -intercept directly from the graph. This method is usually the easiest.
 - Determine the gradient from the graph and using a point that the graph passes through, use a method similar to the previous question to find the equation of the line.
 - Find two points on the graph and use a method similar to the previous question to find the equation of the line. This method will likely take the most amount of time and is the method shown in the solutions.
- (a) From the graph, we can see the graph passes through the points (0,1) and (1,3). We will start by finding the gradient of the line.

$$\begin{aligned}
 m &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{3 - 1}{1 - 0} \\
 &= \frac{2}{1} \\
 &= 2
 \end{aligned}$$

Therefore the gradient of the line is 2 and passes through the point (0,1). We can now find the equation of the line.

$$\begin{aligned}
 y - y_1 &= m(x - x_1) \\
 y - 1 &= 2(x - 0) \\
 y - 1 &= 2x \\
 y - 1 + 1 &= 2x + 1 \\
 y &= 2x + 1
 \end{aligned}$$

- (b) From the graph, we can see that the graph passes through the points (-1,1) and (0,-2). We will start by find the gradient of the line.

$$\begin{aligned}
 m &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{-2 - 1}{0 - -1} \\
 &= \frac{-3}{1} \\
 &= -3
 \end{aligned}$$

Therefore the gradient of the line is -3 and passes through the point (-1,1). We can now find the equation of the line.

$$\begin{aligned}
 y - y_1 &= m(x - x_1) \\
 y - 1 &= -3(x - -1) \\
 y - 1 &= -3(x + 1) \\
 y - 1 &= -3x - 3 \\
 y - 1 + 1 &= -3x - 3 + 1 \\
 y &= -3x - 2
 \end{aligned}$$

4. (a)

$$y = x^2 - 2x - 3$$
$$(a = 1, b = -2, c = -3)$$

The y -intercept is $y = -3$.

We will now find the turning point.

$$\begin{aligned}x &= \frac{-b}{2a} \\&= \frac{-(-2)}{2 \times 1} \\&= \frac{2}{2} \\&= 1\end{aligned}$$

Therefore the x -coordinate of the turning point is $x = 1$. To find the y -coordinate of the turning point, we can substitute $x = 1$ into the original equation.

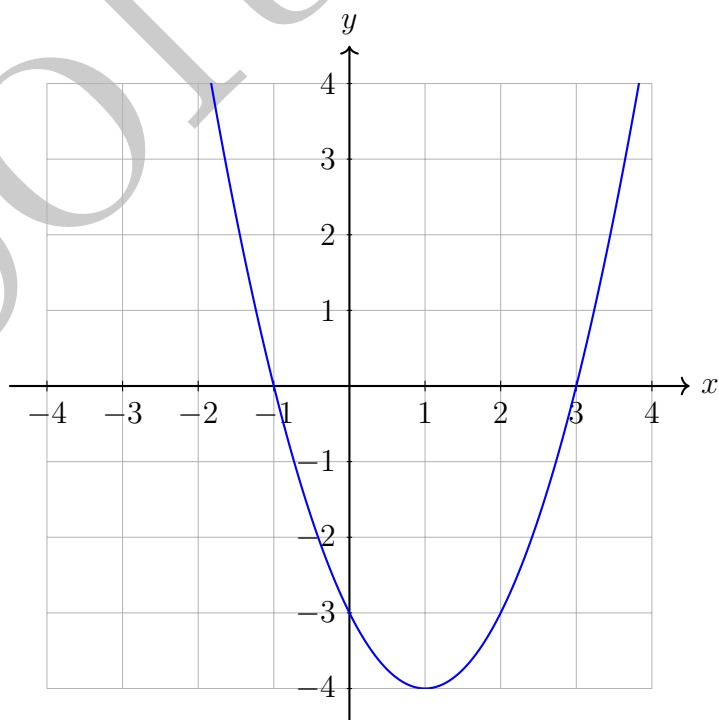
$$\begin{aligned}\text{When } x = 1, y &= 1^2 - 2 \times 1 - 3 \\&= -4\end{aligned}$$

Therefore, the turning point is at $x = 1, y = -4$.

To find x -intercepts, let $y = 0$.

$$\begin{aligned}x^2 - 2x - 3 &= 0 \\(x + 1)(x - 3) &= 0 \\x &= -1, x = 3\end{aligned}$$

Therefore, the x -intercepts are $x = -1$ and $x = 3$.



(b)

$$y = -x^2 + 2x + 8$$
$$(a = -1, b = 2, c = 8)$$

The y -intercept is $y = 8$.

We will now find the turning point.

$$x = \frac{-b}{2a}$$
$$= \frac{-2}{2 \times -1}$$
$$= 1$$

Therefore the x -coordinate of the turning point is $x = 1$. To find the y -coordinate of the turning point, we can substitute $x = 1$ into the original equation.

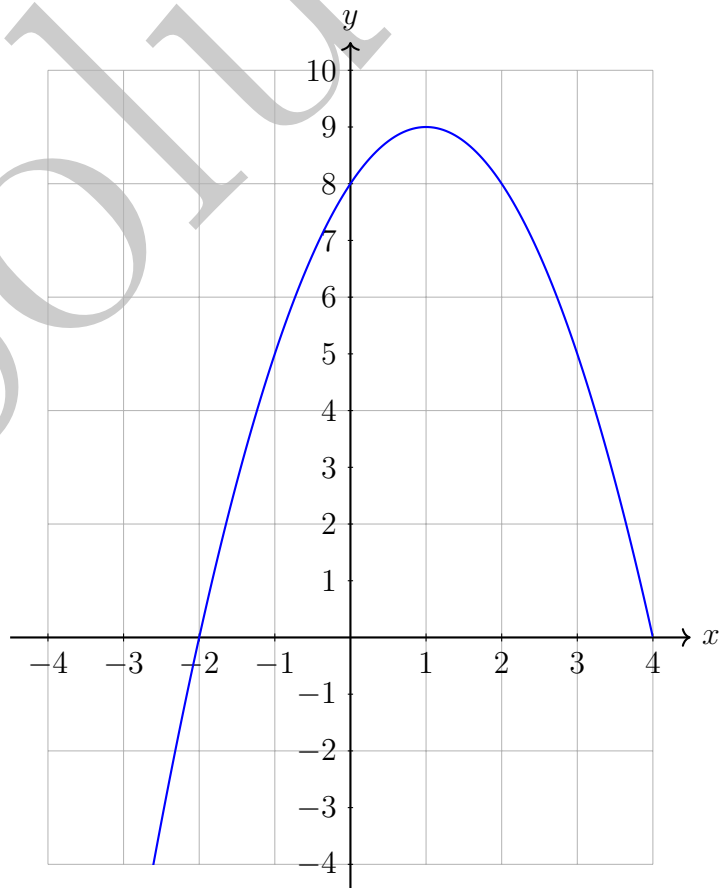
$$\text{When } x = 1, y = -(1)^2 + 2 \times 1 + 8$$
$$= 9$$

Therefore, the turning point is at $x = 1, y = 9$.

To find x -intercepts, let $y = 0$.

$$-x^2 + 2x + 8 = 0$$
$$(-x + 4)(x + 2) = 0$$
$$x = 4, x = -2$$

Therefore the x -intercepts are $x = -2$ and at $x = 4$.



(c)

$$y = x^2 - 9$$

For this question, it may make it easier to first rewrite the quadratic in the form $y = ax^2 + bx + c$.

$$\begin{aligned} y &= x^2 + 0x - 9 \\ (a = 1, b = 0, c = -9) \end{aligned}$$

The y -intercept is at $y = -9$.

We will now find the turning point.

$$\begin{aligned} x &= \frac{-b}{2a} \\ &= \frac{0}{2 \times 1} \\ &= 0 \end{aligned}$$

Therefore the x -coordinate of the turning point is $x = 0$. To find the y -coordinate of the turning point, we can substitute $x = 0$ into the original equation.

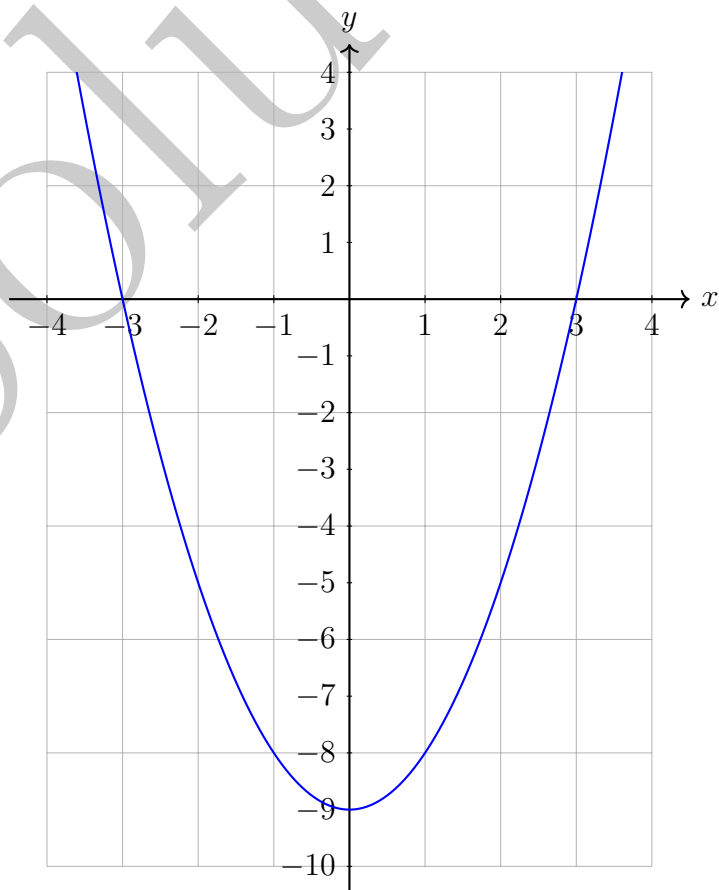
$$\begin{aligned} \text{When } x = 0, y &= 0^2 + 0 \times 0 - 9 \\ &= -9 \end{aligned}$$

Therefore, the turning point is at $x = 0, y = -9$.

To find x -intercepts, let $y = 0$.

$$\begin{aligned} x^2 - 9 &= 0 \\ (x + 3)(x - 3) &= 0 \\ x &= -3, x = 3 \end{aligned}$$

Therefore the x -intercepts are $x = -3$ and $x = 3$.



(d)

$$y = -2x^2 - 12x - 10$$
$$(a = -2, b = -12, c = -10)$$

The y -intercept is at $y = -10$.

We will now find the turning point.

$$\begin{aligned}x &= \frac{-b}{2a} \\&= \frac{-(-12)}{2 \times -2} \\&= \frac{12}{-4} \\&= -3\end{aligned}$$

Therefore the x -coordinate of the turning point is $x = -3$. To find the y -coordinate of the turning point, we can substitute $x = -3$ into the original equation.

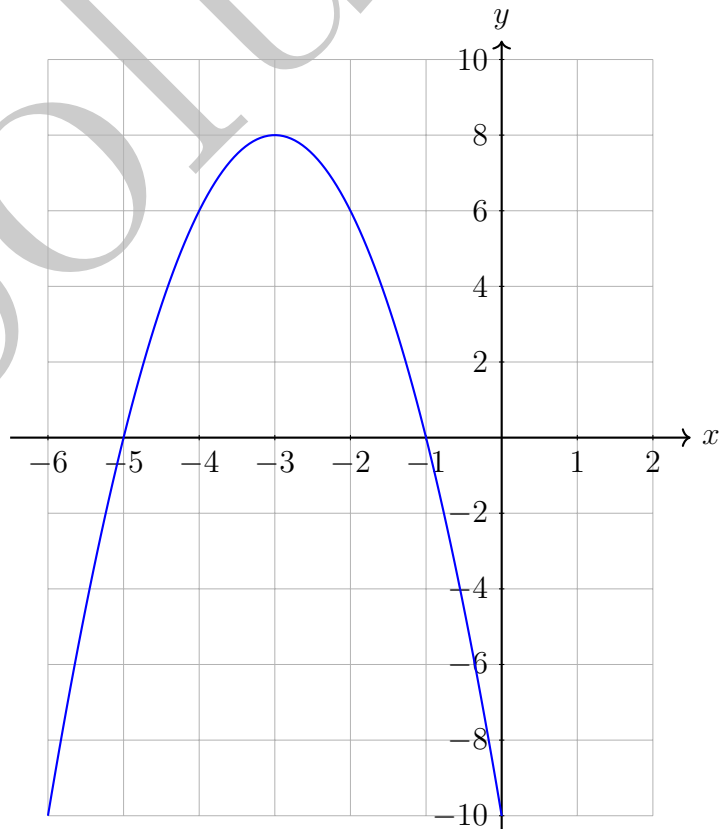
$$\begin{aligned}\text{When } x = -3, y &= -2 \times (-3)^2 - 12 \times -3 - 10 \\&= 8\end{aligned}$$

Therefore, the turning point is at $x = -3, y = 8$.

To find x -intercepts, let $y = 0$.

$$\begin{aligned}-2x^2 - 12x - 10 &= 0 \\-2(x^2 + 6x + 5) &= 0 \\-2(x + 5)(x + 1) &= 0 \\x = -5, x &= -1\end{aligned}$$

Therefore the x -intercepts are $x = -5$ and $x = -1$.



(e)

$$y = x^2 - 8x + 16$$
$$(a = 1, b = -8, c = 16)$$

The y -intercept is at $y = 16$.

We will now find the turning point.

$$\begin{aligned}x &= \frac{-b}{2a} \\&= \frac{-(-8)}{2 \times 1} \\&= \frac{8}{2} \\&= 4\end{aligned}$$

Therefore the x -coordinate of the turning point is $x = 4$. To find the y -coordinate of the turning point, we can substitute $x = 4$ into the original equation.

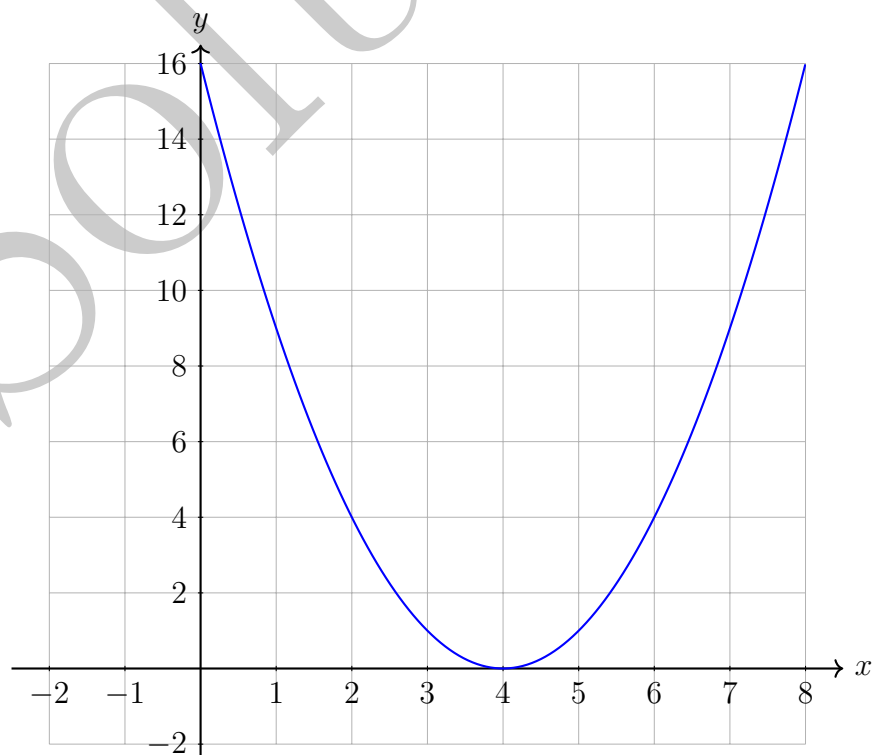
$$\begin{aligned}\text{When } x = 4, y &= 4^2 - 8 \times 4 + 16 \\&= 0\end{aligned}$$

Therefore, the turning point is at $x = 4, y = 0$.

To find x -intercepts, let $y = 0$.

$$\begin{aligned}x^2 - 8x + 16 &= 0 \\(x - 4)(x - 4) &= 0 \\x = 4, x &= 4\end{aligned}$$

Therefore the x -intercept is $x = 4$. Note that this quadratic only has one unique x -intercept.



(f)

$$y = x^2 - 4x + 7$$
$$(a = 1, b = -4, c = 7)$$

The y -intercept is at $y = 7$.

We will now find the turning point.

$$\begin{aligned}x &= \frac{-b}{2a} \\&= \frac{-(-4)}{2 \times 1} \\&= \frac{4}{2} \\&= 2\end{aligned}$$

Therefore the x -coordinate of the turning point is $x = 2$. To find the y -coordinate of the turning point, we can substitute $x = 2$ into the original equation.

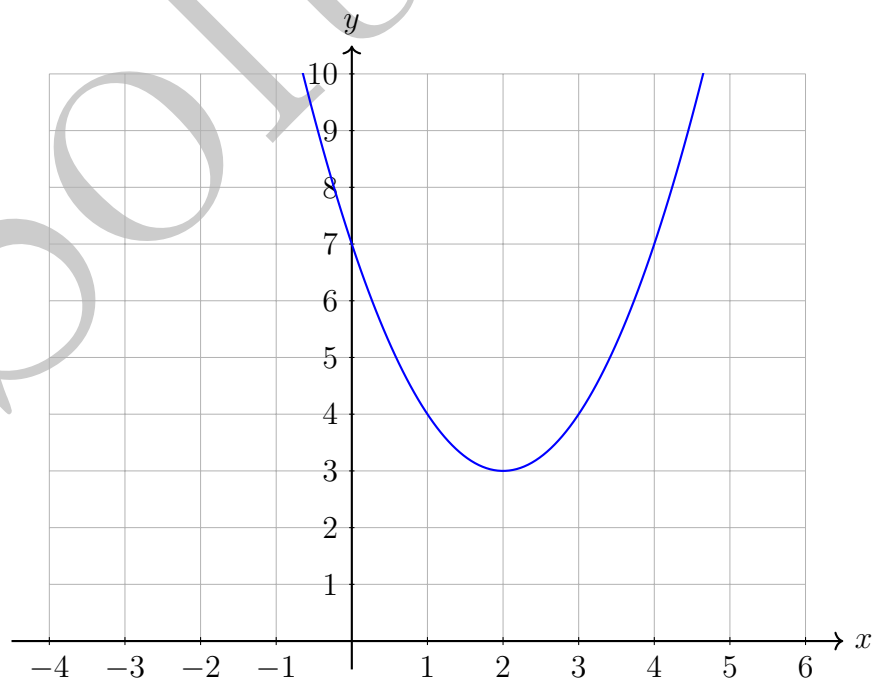
$$\begin{aligned}\text{When } x = 2, y &= 2^2 - 4 \times 2 + 7 \\&= 3\end{aligned}$$

Therefore, the turning point is at $x = 2, y = 3$.

To find x -intercepts, let $y = 0$.

$$x^2 - 2x - 3 = 0$$

However, this quadratic has no (real) solutions. Therefore, there are no x -intercepts. (You can try to solve this quadratic using the quadratic formula, but it will involve the square root of a negative number).



5. (a)

$$3x + 2y = 7 \quad (1)$$

$$5x - y = 3 \quad (2)$$

Rearranging Equation (2) gives

$$5x - y = 3$$

$$5x - y + y = 3 + y$$

$$5x = 3 + y$$

$$5x - 3 = 3 - 3 + y$$

$$y = 5x - 3$$

Substituting $y = 5x - 3$ into Equation (1) gives

$$3x + 2(5x - 3) = 7$$

$$3x + 10x - 6 = 7$$

$$13x - 6 = 7$$

$$13x = 13$$

$$x = 1$$

Substituting $x = 1$ into $y = 5x - 3$ gives

$$y = 5 \times 1 - 3$$

$$= 5 - 3$$

$$= 2$$

Therefore, $x = 1$, $y = 2$.

(b)

$$3x + 5y = -5 \quad (1)$$

$$4x - y = 24 \quad (2)$$

Rearranging Equation (2) gives

$$4x - y = 24$$

$$4x - y + y = 24 + y$$

$$4x = 24 + y$$

$$4x - 24 = 24 - 24 + y$$

$$y = 4x - 24$$

Substituting $y = 4x - 24$ into Equation (1) gives

$$3x + 5(4x - 24) = -5$$

$$3x + 20x - 120 = -5$$

$$23x - 120 = -5$$

$$23x = 115$$

$$x = 5$$

Substituting $x = 5$ into $y = 4x - 24$ gives

$$\begin{aligned}y &= 4 \times 5 - 24 \\&= 20 - 24 \\&= -4\end{aligned}$$

Therefore $x = 5$, $y = -4$.

(c)

$$4x + 3y = -9 \quad (1)$$

$$3x + 2y = -7 \quad (2)$$

Rearranging Equation (2) gives

$$\begin{aligned}3x + 2y &= -7 \\3x - 3x + 2y &= -3x - 7 \\2y &= -3x - 7 \\\frac{2y}{2} &= \frac{-3x - 7}{2} \\y &= \frac{-3x - 7}{2}\end{aligned}$$

Substituting $y = \frac{-3x-7}{2}$ into Equation (1) gives

$$\begin{aligned}4x + 3 \times \left(\frac{-3x - 7}{2} \right) &= -9 \\4x + \frac{-9x - 21}{2} &= -9\end{aligned}$$

Multiplying through by 2 gives

$$\begin{aligned}8x - 9x - 21 &= -18 \\-x - 21 &= -18 \\-x &= 3 \\x &= -3\end{aligned}$$

Substituting $x = -3$ into $y = \frac{-3x-7}{2}$ gives

$$\begin{aligned}y &= \frac{-3 \times -3 - 7}{2} \\&= \frac{9 - 7}{2} \\&= \frac{2}{2} \\&= 1\end{aligned}$$

Therefore, $x = -3$, $y = 1$.

(d)

$$5x - 4y = 10 \quad (1)$$

$$4x - 3y = 7 \quad (2)$$

Rearranging Equation (2) gives

$$\begin{aligned} 4x - 3y &= 7 \\ 4x - 3y + 3y &= 3y + 7 \\ 4x &= 3y + 7 \\ 4x - 7 &= 3y + 7 - 7 \\ 4x - 7 &= 3y \\ \frac{4x - 7}{3} &= \frac{3y}{3} \\ y &= \frac{4x - 7}{3} \end{aligned}$$

Substituting $y = \frac{4x-7}{3}$ into Equation (1) gives

$$\begin{aligned} 5x - 4 \times \left(\frac{4x - 7}{3} \right) &= 10 \\ 5x - \frac{16x - 28}{3} &= 10 \end{aligned}$$

Multiplying through by 3 gives

$$\begin{aligned} 15x - (16x - 28) &= 30 \\ 15x - 16x + 28 &= 30 \\ -x + 28 &= 30 \\ -x &= 2 \\ x &= -2 \end{aligned}$$

Substituting $x = -2$ into $y = \frac{4x-7}{3}$ gives

$$\begin{aligned} y &= \frac{4 \times -2 - 7}{3} \\ &= \frac{-8 - 7}{3} \\ &= \frac{-15}{3} \\ &= -5 \end{aligned}$$

Therefore $x = -2$, $y = -5$.

(e)

$$5x + 2y = -4 \quad (1)$$

$$7x + 3y = -6 \quad (2)$$

Rearranging Equation (1) gives

$$5x + 2y = -4$$

$$5x - 5x + 2y = -5x - 4$$

$$2y = -5x - 4$$

$$\frac{2y}{2} = \frac{-5x - 4}{2} \quad y = \frac{-5x - 4}{2}$$

Substituting $y = \frac{-5x-4}{2}$ into Equation (2) gives

$$7x + 3 \times \left(\frac{-5x - 4}{2} \right) = -6$$

Multiplying through by 2 gives

$$14x + 3(-5x - 4) = -12$$

$$14x - 15x - 12 = -12$$

$$-x - 12 = -12$$

$$-x - 12 + 12 = -12 + 12$$

$$-1x = 0$$

$$\frac{-1x}{-1} = \frac{0}{-1}$$

$$x = 0$$

Substituting $x = 0$ into $y = \frac{-5x-4}{2}$ gives

$$y = \frac{-5 \times 0 - 4}{2}$$

$$= \frac{0 - 4}{2}$$

$$= \frac{-4}{2}$$

$$= -2$$

Therefore $x = 0, y = -2$.

(f)

$$4x - 3y = -19 \quad (1)$$

$$10x - 2y = -42 \quad (2)$$

Rearranging Equation (2) gives

$$y = \frac{10x + 42}{2}$$

$$y = \frac{2(5x + 21)}{2}$$

$$y = 5x + 21$$

Substituting $y = 5x + 21$ into Equation (1) gives

$$\begin{aligned}4x - 3(5x + 21) &= -19 \\4x - 15x - 63 &= -19 \\-11x - 63 &= -19 \\-11x - 63 + 63 &= -19 + 63 \\-11x &= 44 \\\frac{-11x}{-11} &= \frac{44}{-11} \\x &= -4\end{aligned}$$

Substituting $x = -4$ into $y = 5x + 21$ gives

$$\begin{aligned}y &= 5 \times -4 + 21 \\&= -20 + 21 \\&= 1\end{aligned}$$

Therefore $x = -4$, $y = 1$.